14 **(a)** Use the laws of logarithms to show that $\log_{10}200 - \log_{10}20$ is equal to 1. **[2]**

The first three terms of a sequence are $\log_{10}20, \log_{10}200, \log_{10}2000$.

(b) Show that the sequence is arithmetic. **[2]**

(c) Find the exact value of the sum of the first 50 terms of this sequence. **[2]**